

Study Questions: Perception

1. Describe the predictive coding architecture. How do deep pyramidal cells and superficial pyramidal cells implement predictions and prediction errors?
2. How did Rao and Ballard (1999) demonstrate predictive coding in the visual cortex?
3. What is sensory attenuation? Why can't you tickle yourself?
4. How does the cerebellum contribute to sensory attenuation through efference copies?
5. Explain how failed sensory attenuation could lead to auditory hallucinations in schizophrenia.
6. What did Powers et al. (2017) demonstrate about hallucination-prone individuals and Pavlovian conditioning?
7. How does Active Inference explain visual hallucinations in Charles Bonnet syndrome?
8. What is the role of precision in distinguishing between normal perception and hallucination?
9. How do attention and precision relate to the gain control of prediction error signals?
10. Compare "bottom-up" and "top-down" processing in the predictive coding hierarchy. Why is "bottom-up" a misleading term?
11. How does mismatch negativity (MMN) in auditory cortex provide evidence for predictive coding?
12. What is the "Bayesian brain hypothesis"? How does predictive coding provide a neural implementation?
13. How does the concept of "explaining away" (suppression of prediction error by accurate predictions) work in visual perception?
14. What experimental evidence supports the claim that V1 neurons respond to prediction errors, not raw sensory input?
15. How does binocular rivalry (competition between different images presented to each eye) relate to predictive coding?
16. What is the role of NMDA receptors in implementing prediction error signaling at the cortical level?
17. How does Active Inference's account of perception differ from the "feature detector" model (Hubel & Wiesel)?

18. Compare top-down hallucination (overweighting predictions) with bottom-up hyperesthesia (overweighting sensory input). Give a clinical example of each.
19. How might psychedelic drugs (which increase neural entropy) alter the predictive coding hierarchy?
20. Design an experiment that could test whether visual hallucinations in Charles Bonnet syndrome are suppressed by increasing bottom-up sensory input (e.g., brighter lighting, more visual texture).